

## # Unit vectors

Q. Select the unit vector in the direction of  $\vec{v} = (-5, -4)$ .  
choose 1 answer:

Sol

The unit vector in the direction of a non zero vector  $\vec{v}$  is given by

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|}$$

Here,  $\vec{v} = (-5, -4)$ , so

$$\|\vec{v}\| = \sqrt{(-5)^2 + (-4)^2} = \sqrt{25+16} = \sqrt{41}$$

Therefore

$$\hat{v} = \left( \frac{-5}{\sqrt{41}}, \frac{-4}{\sqrt{41}} \right)$$

Hence, choice D.

Q. Find the unit vector in the direction of  $\vec{v} = (6, 6)$ .

Sol  $\hat{v} = \frac{\vec{v}}{\|\vec{v}\|}$

Here,  $\vec{v} = (6, 6)$ , so

$$\|\vec{v}\| = \sqrt{6^2 + 6^2} = \sqrt{36+36} = \sqrt{72} = 6\sqrt{2}$$

Therefore

$$\begin{aligned} \hat{v} &= \left( \frac{6}{6\sqrt{2}}, \frac{6}{6\sqrt{2}} \right) \\ &= \left( \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) \end{aligned}$$

Q. select the unit vector in the direction of  $\vec{v} = (-1, -3)$ .

Sol  $\hat{v} = \frac{\vec{v}}{\|\vec{v}\|}$

Here,  $\vec{v} = (-1, -3)$ , so

$$\|\vec{v}\| = \sqrt{(-1)^2 + (-3)^2} = \sqrt{1+9} = \sqrt{10}$$

Therefore

$$\hat{v} = \left( \frac{-1}{\sqrt{10}}, \frac{-3}{\sqrt{10}} \right)$$

Q. Find the unit vectors in the direction of  $\vec{v} = (7, -3)$ .

Sol:  $\hat{v} = \frac{\vec{v}}{\|\vec{v}\|}$

Here,  $\|\vec{v}\| = \sqrt{(7)^2 + (-3)^2} = \sqrt{49+9} = \sqrt{58}$

Therefore,  $\hat{v} = \left( \frac{7}{\sqrt{58}}, -\frac{3}{\sqrt{58}} \right)$